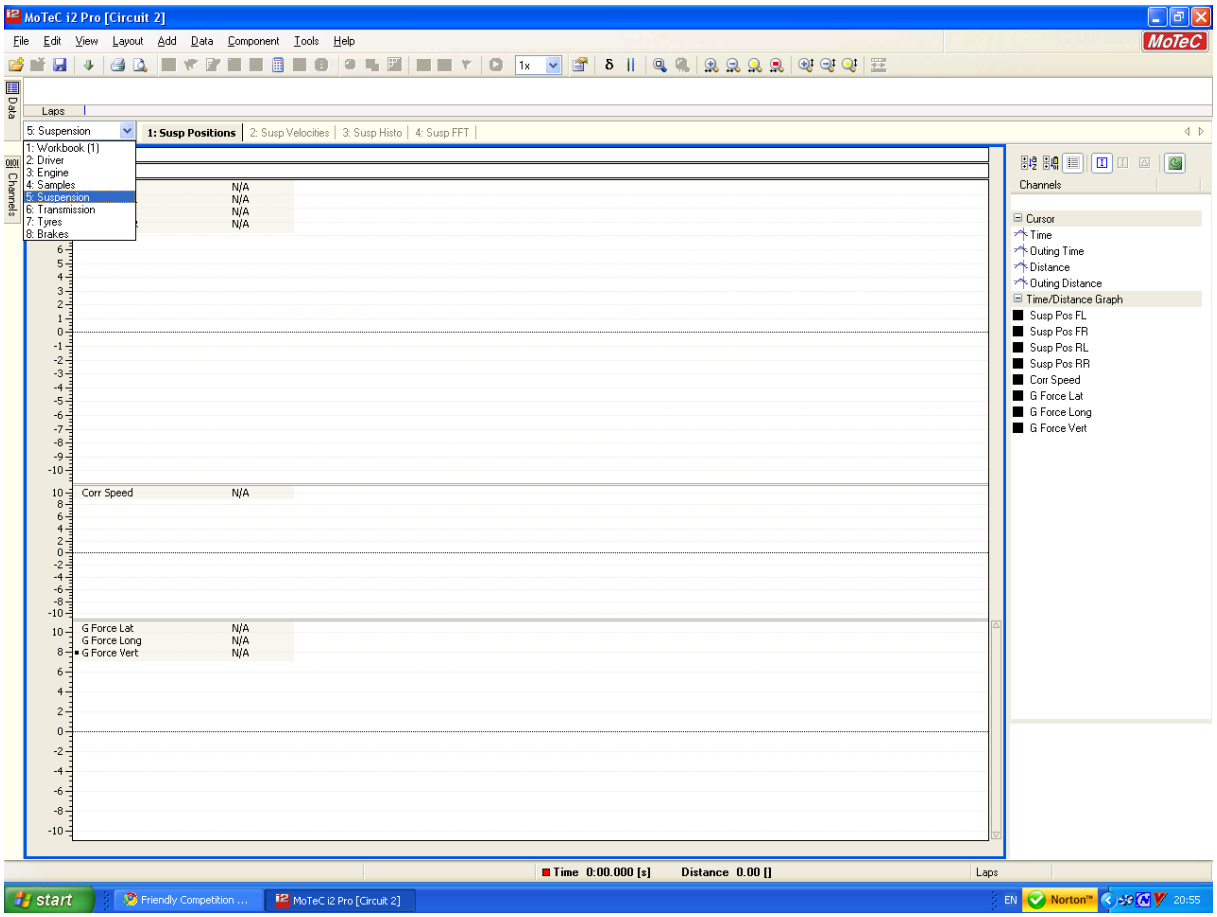
How to Set Up a Suspension (Damper) Velocity Graph using Time as the Y Axis

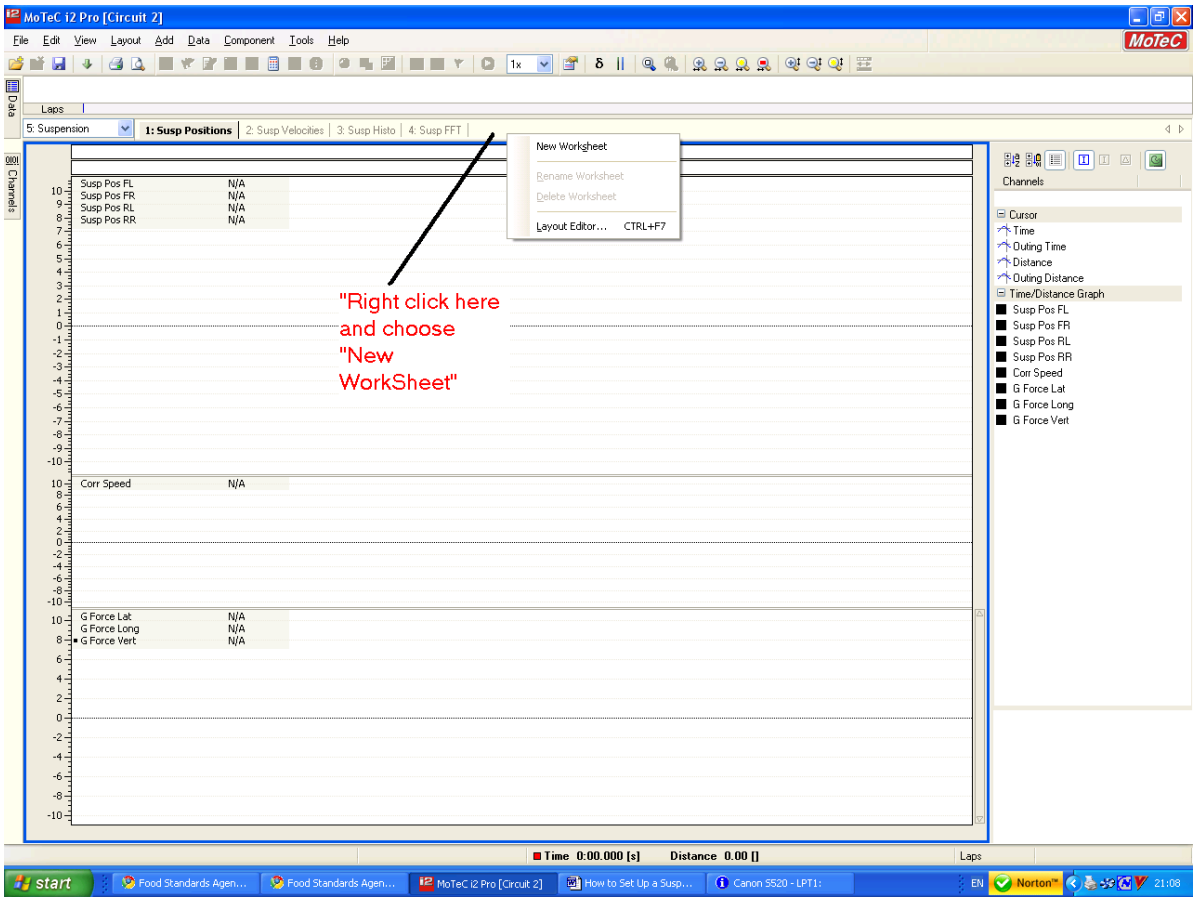
By Allan Paterson
(Encouraged by Animal Ed, Spad & Eagleone)

I've written this because of several posts (In Spad's "My MoTeC Project and Explanation" thread in the GTR Evolution Forum at www.nogripracing.com) wondering how it was done, including post #111, where I asked the question myself sometime ago.

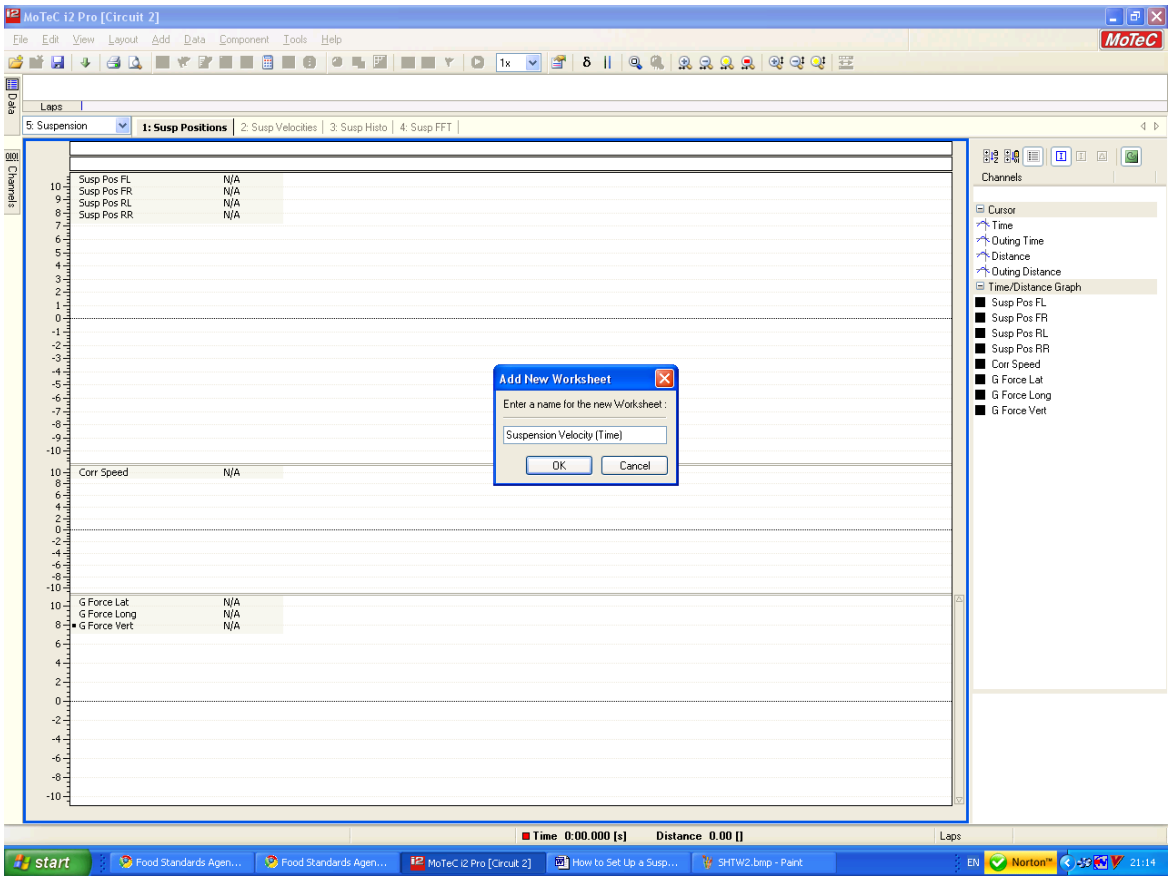
First the logical place to put this new "Worksheet" is in the "Suspension" tab – however you need to open a logged data file for all options to work;



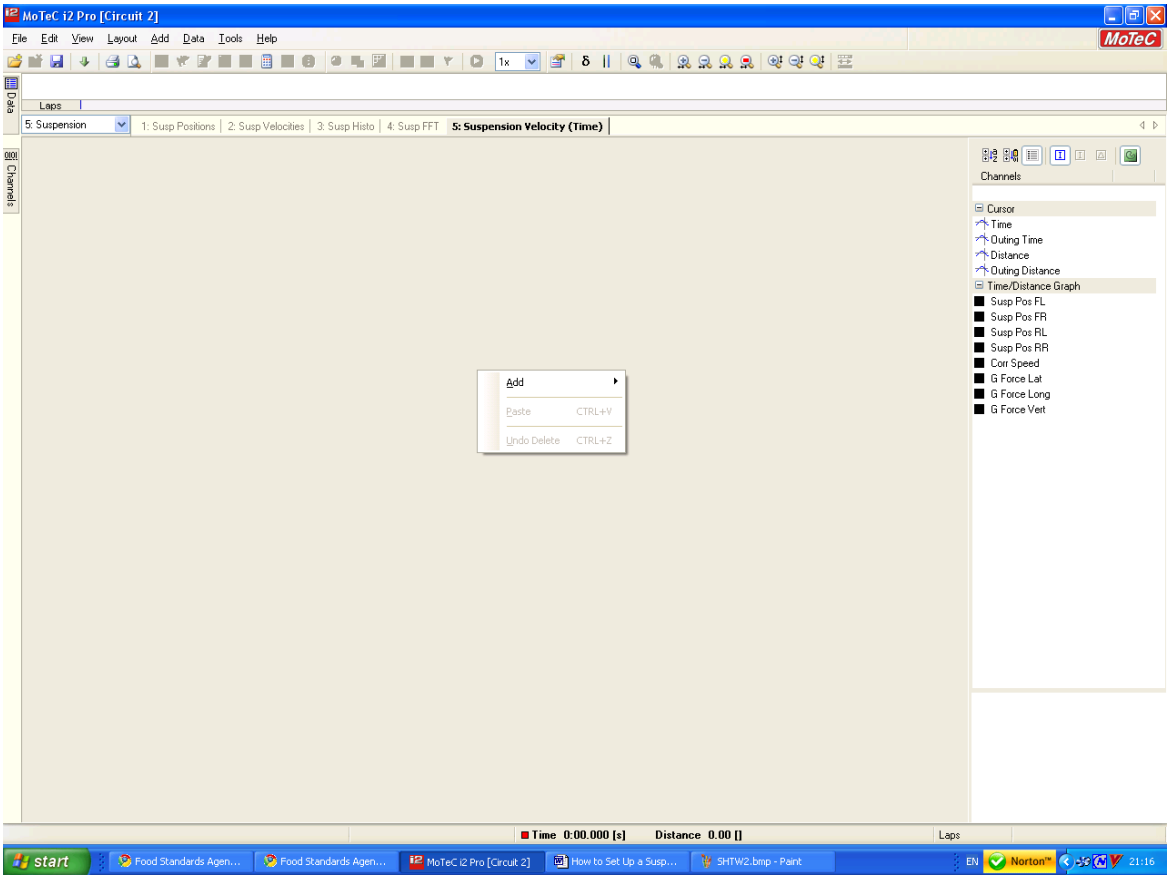
Now “Right click” in the blank area to the right of the worksheet headings;



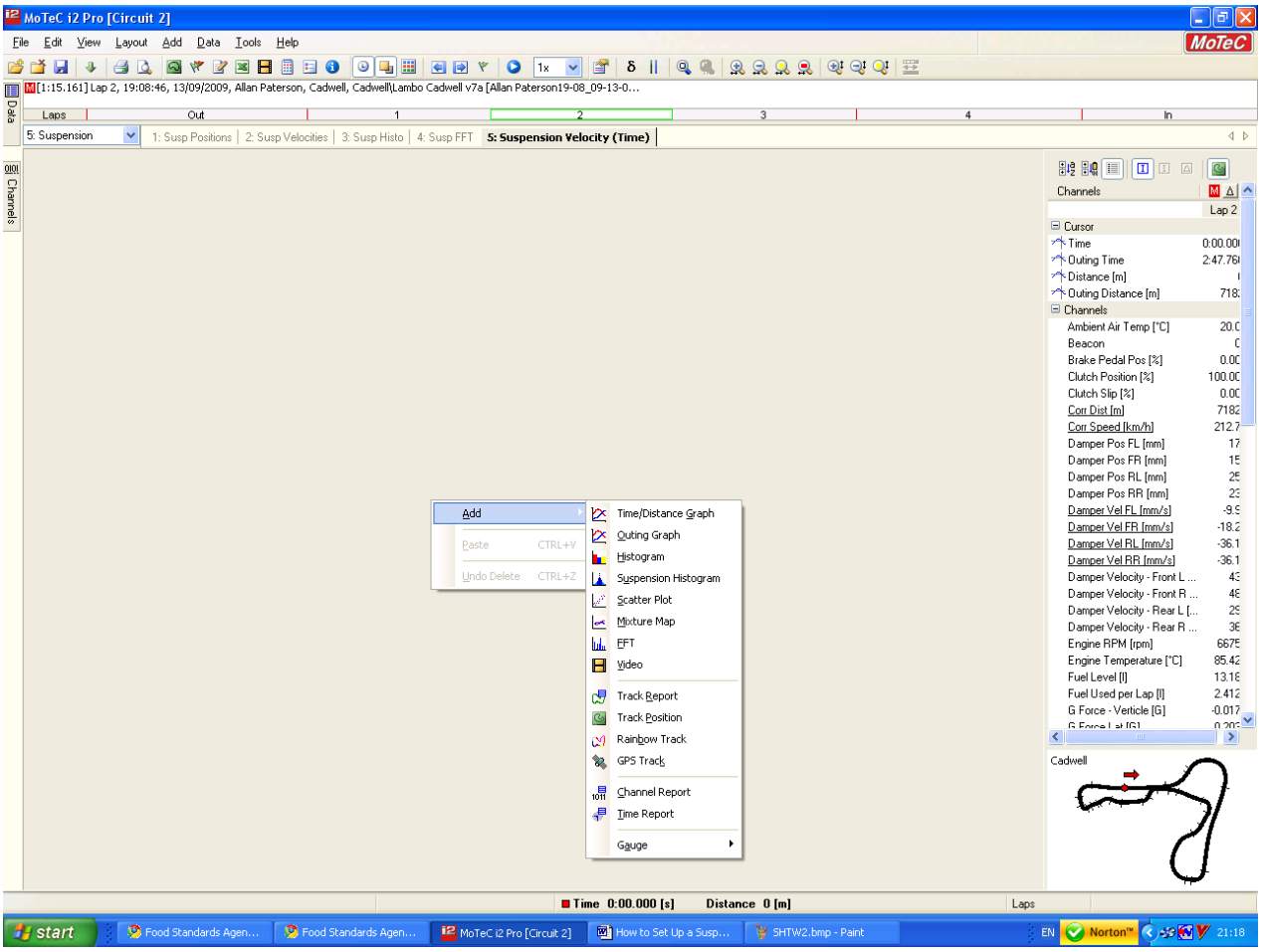
Call your new worksheet what you want, I’ve called mine Suspension Velocity (Time)



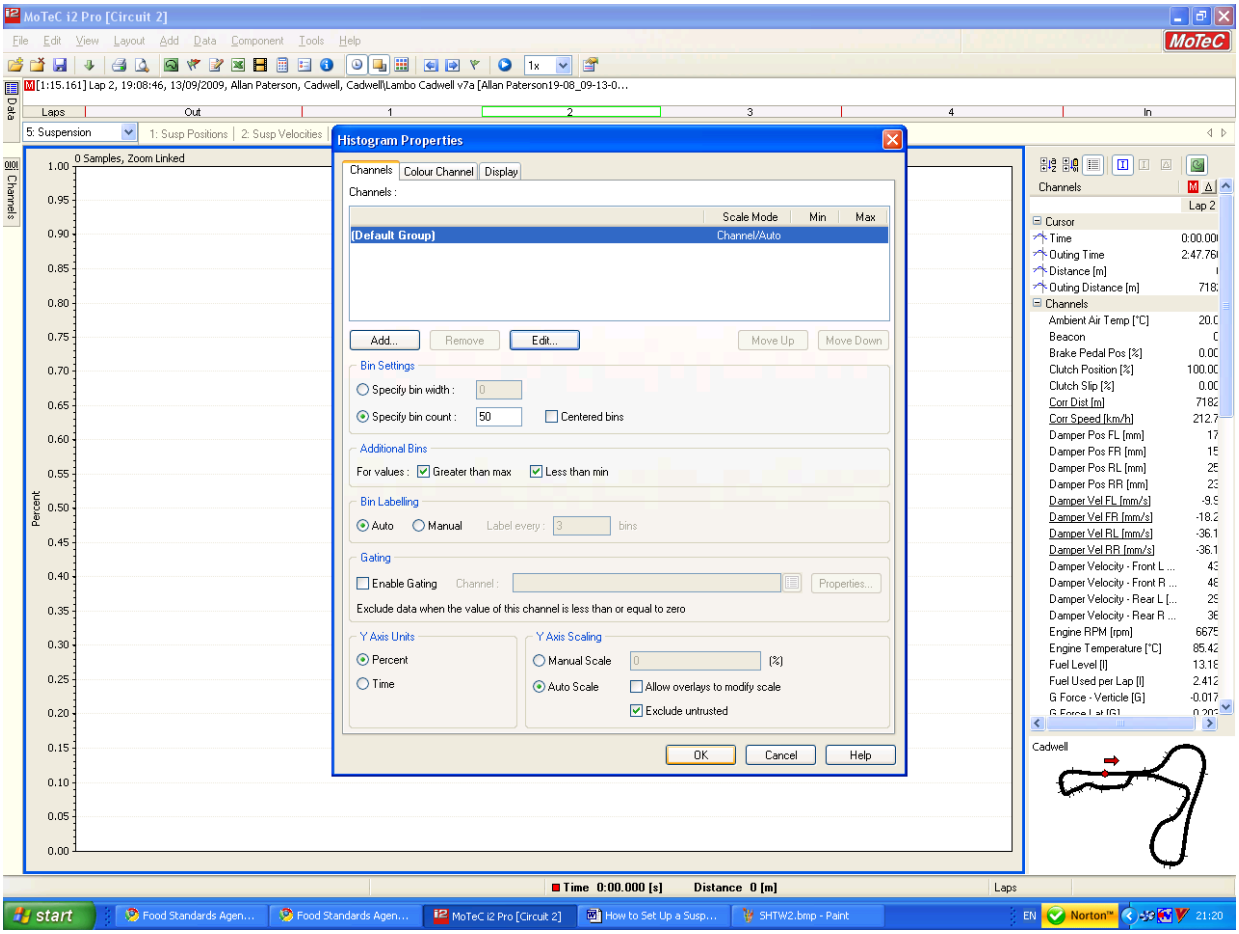
This following section will need to be repeated 4 times to create Histograms for each corner, now "Right Click in the blank area of your new Worksheet and you get this dialog box;



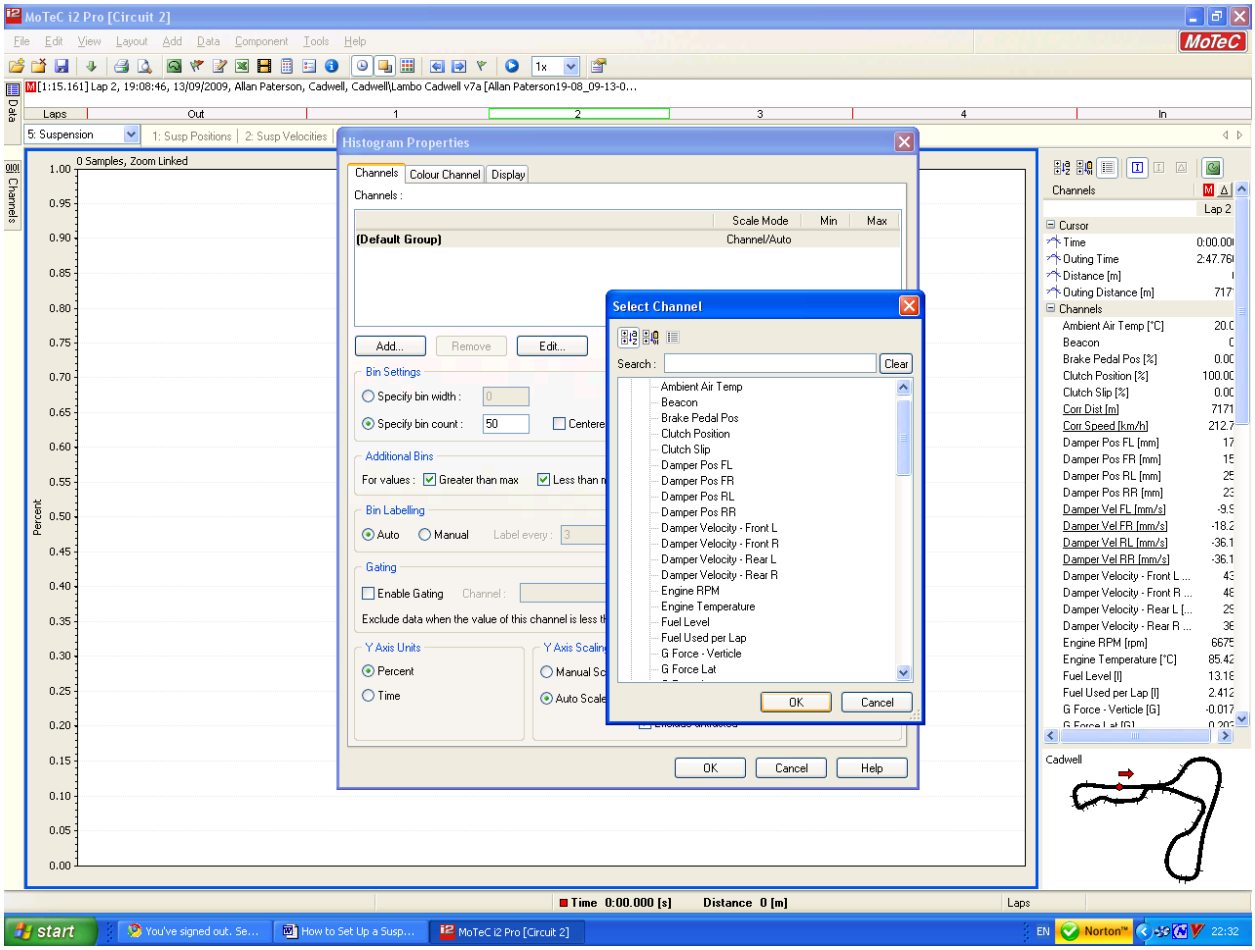
Click on Add;



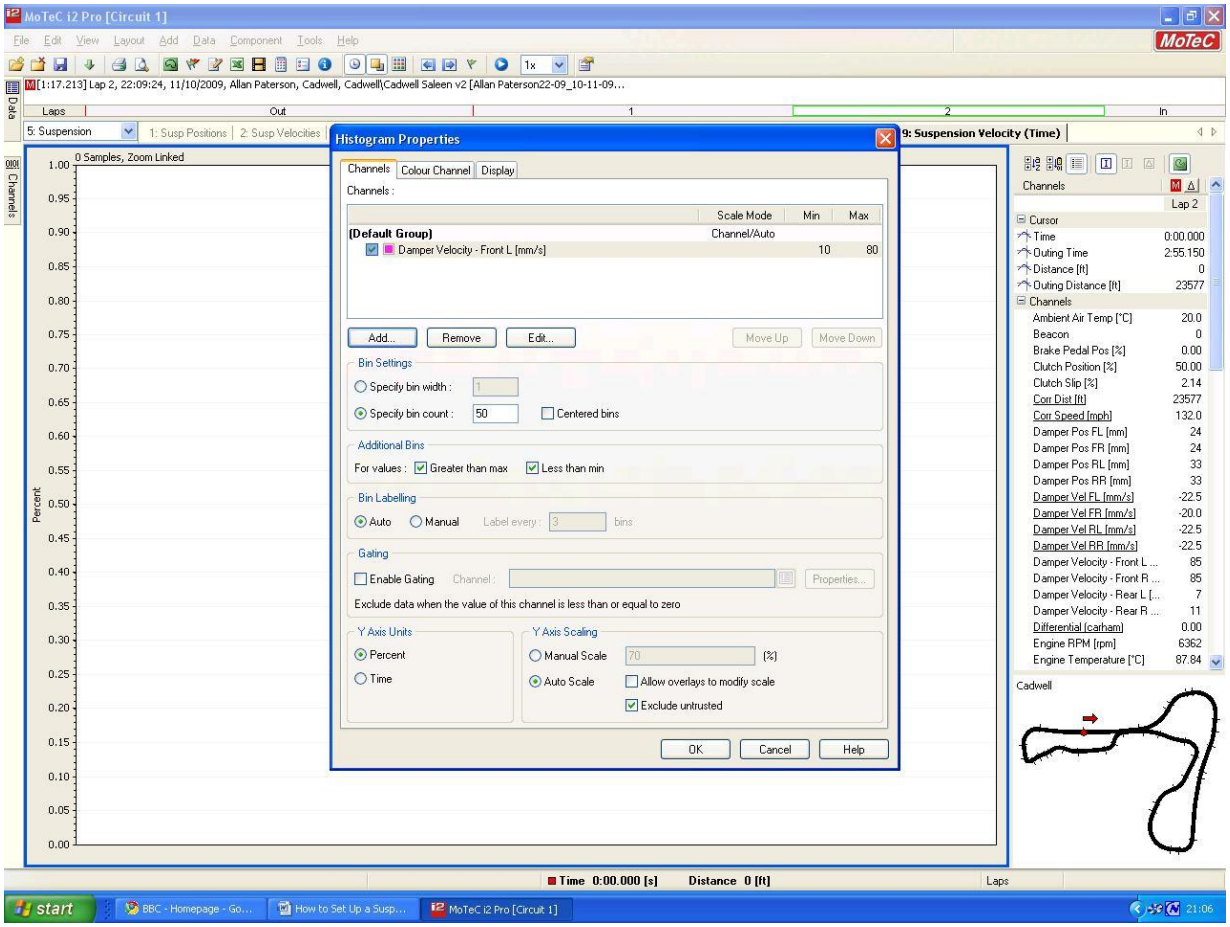
Choose “Histogram”, third one down, **do not choose** “Suspension Histogram”
You will get a screen that looks like this;



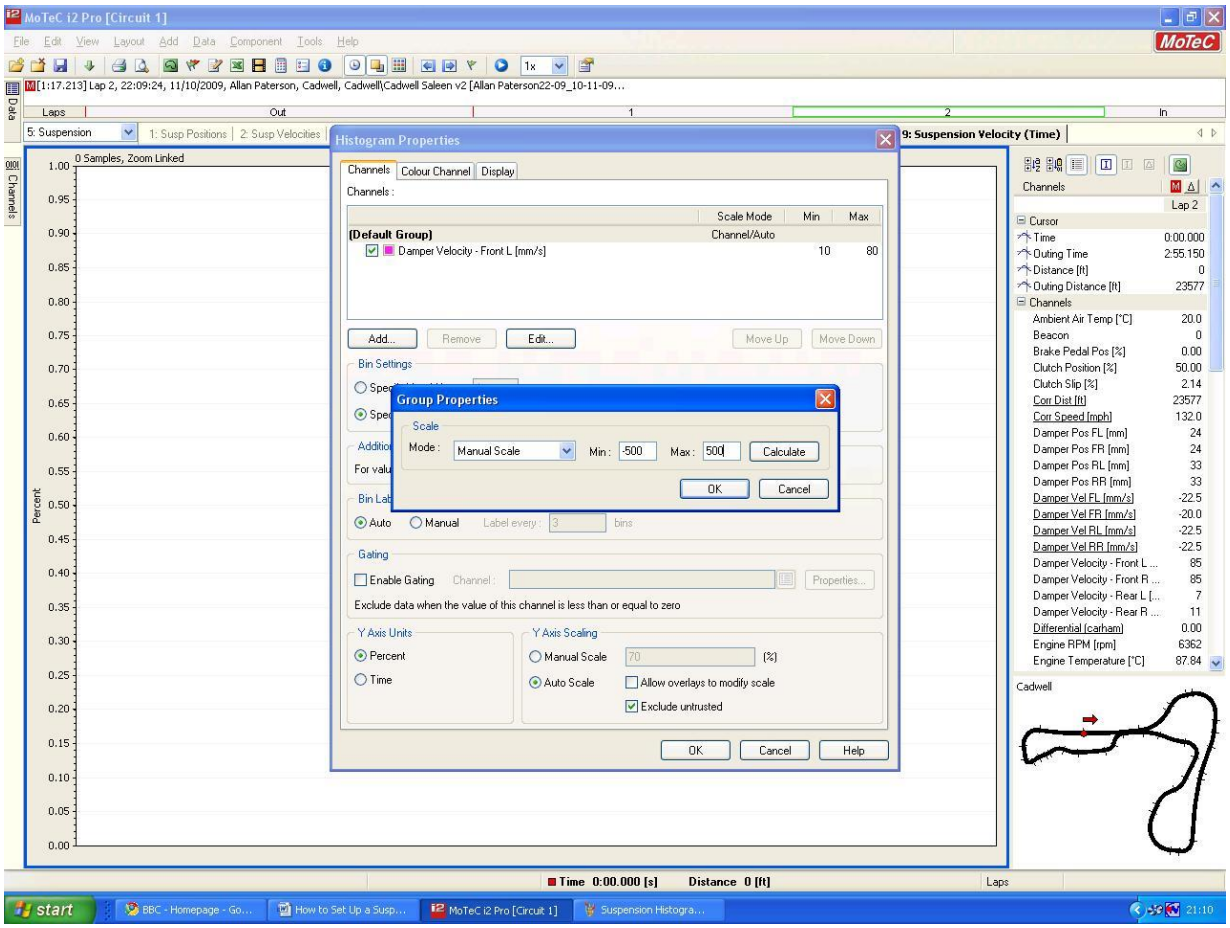
Click on the “Add” button to bring up the select Channel menu;



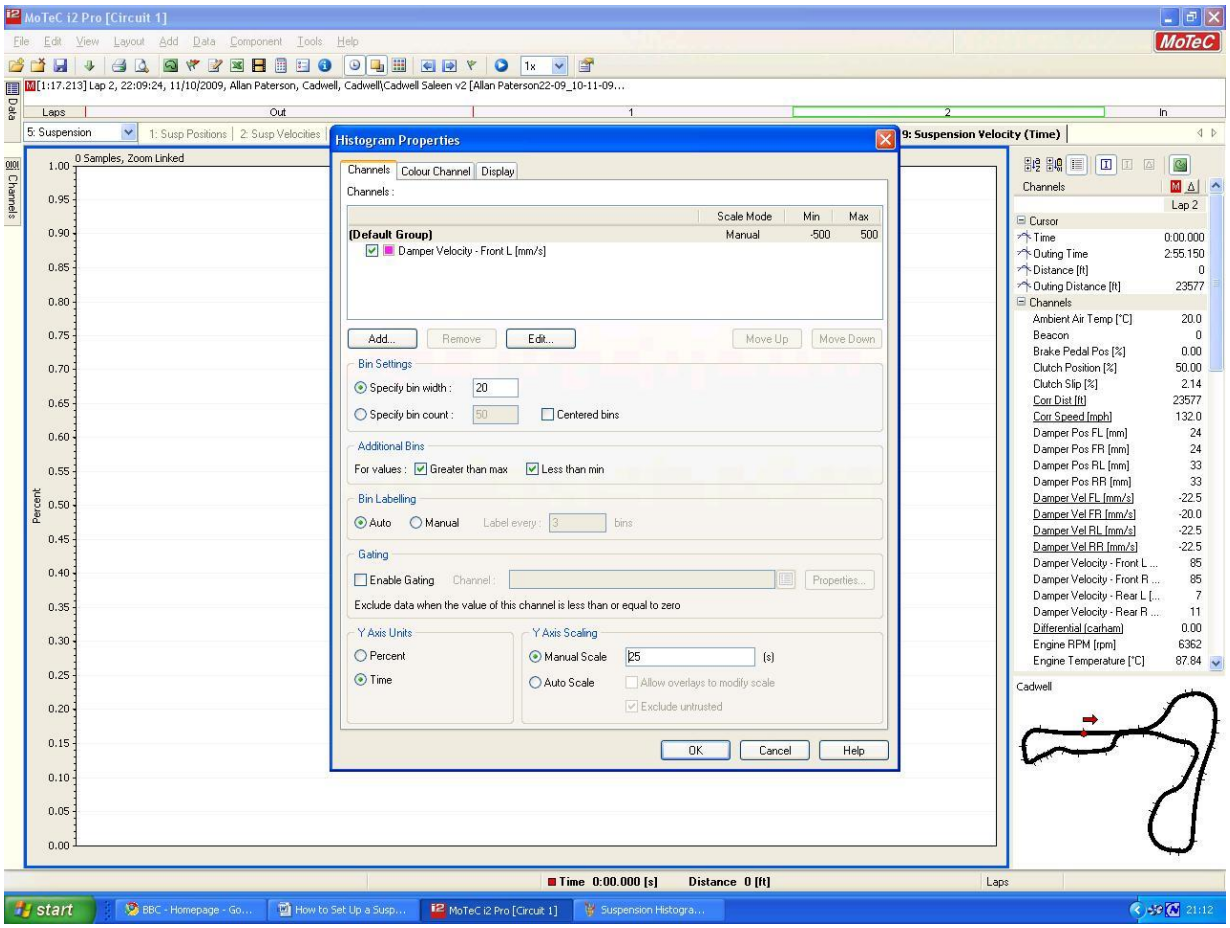
Add the first of the four Damper Velocity Channels, start with FL, and each time you repeat this step chose the next corner in turn FR, RL, RR;



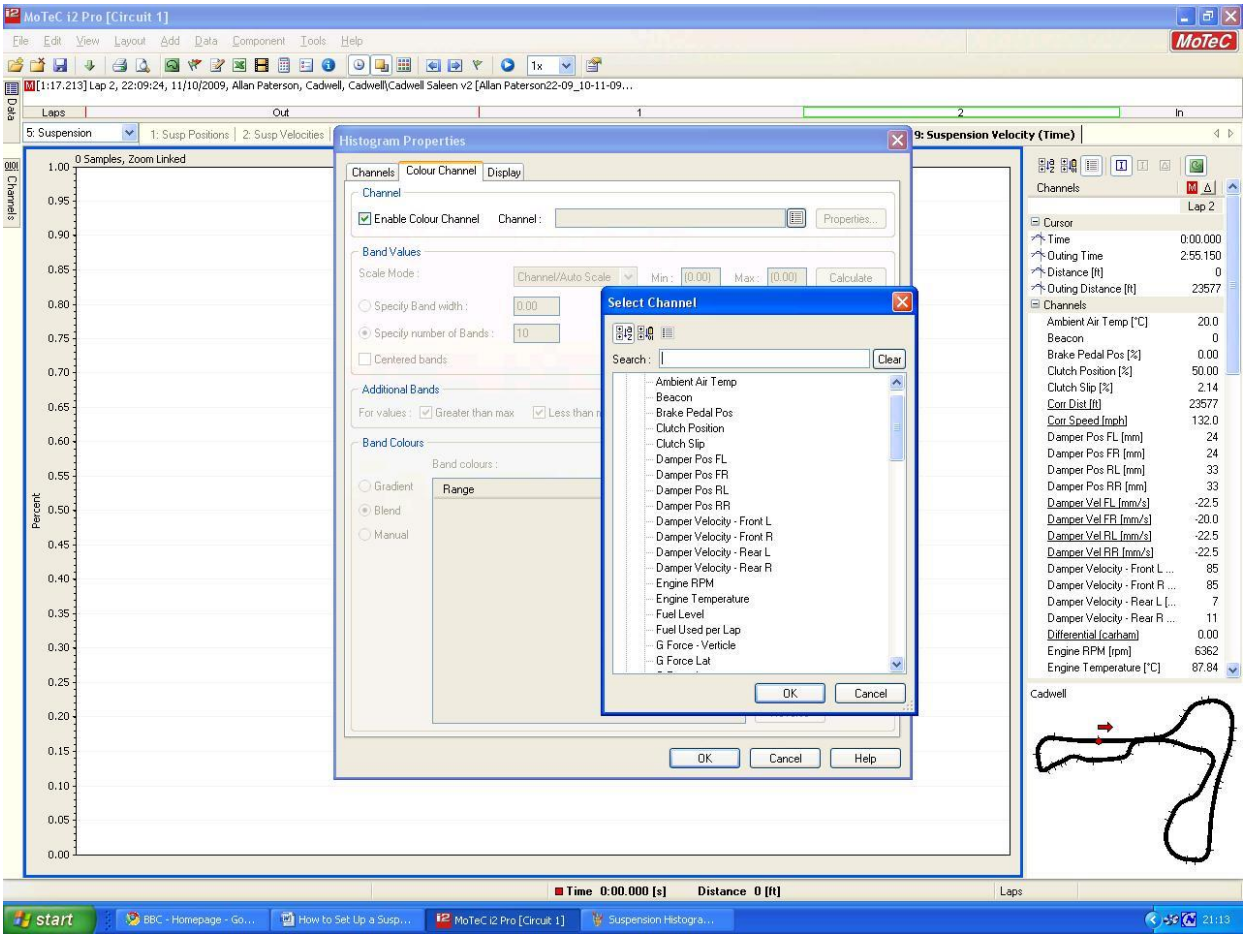
Before clicking on Okay highlight the Top line "[Default Group] and click "Edit" changing the Group properties to Manual Scale and making Minimum –500 and Maximum 500, Click Okay;



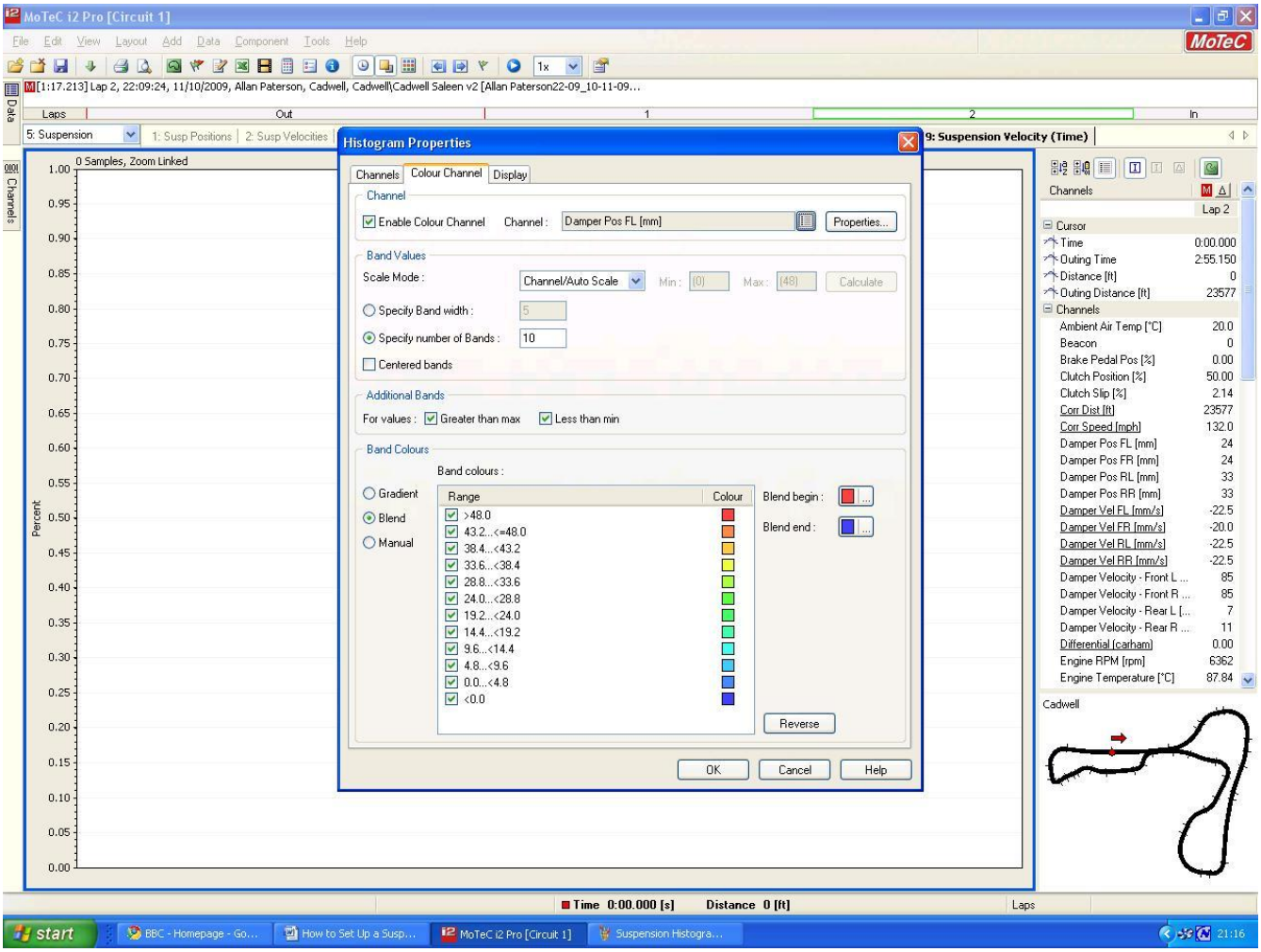
Then change “Bin Settings” to Specify Bin Width, make it 20, Change Y Axis Units to “Time” bottom left hand side and Y axis scaling to Manual Scale and set this to 25, Click Okay;



Now click the “Colour Channel Tab at the top and tick the Enable Colour Channel box;

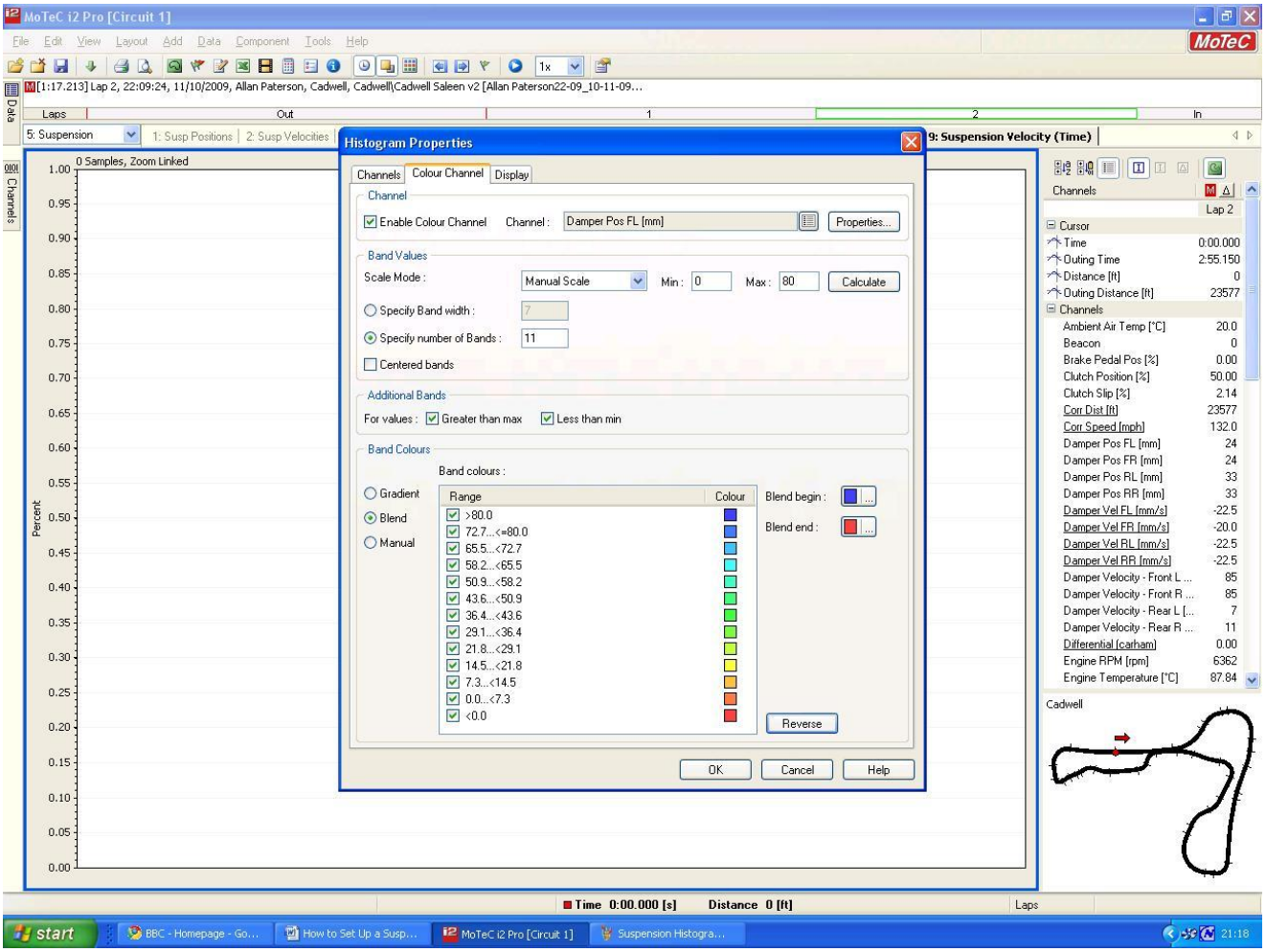


Now click the box at the end of the "Channel" long box, you will get a "Select Channel" box as per the above screen shot; Then Add Damper Position – FL (and change this each time you repeat this step to be the same damper position as the damper velocity you chose in step 1)

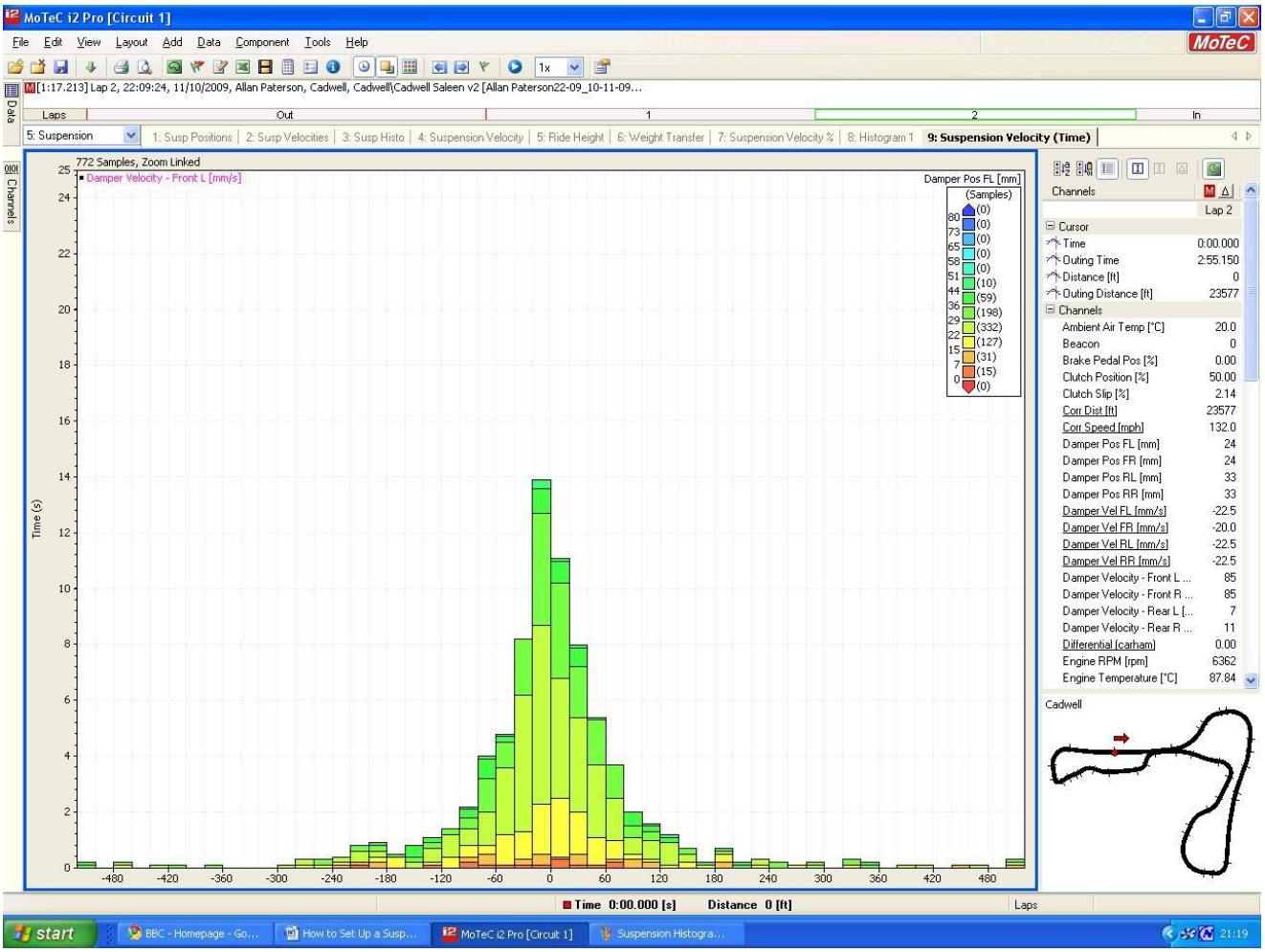


On the Histogram Properties box under "Band Values" change "Scale Mode" to Manual Scale and set Minimum to 0 and Maximum to 80. Click on Specify Number of Bands and set that to 11. Finally Click "Reverse" next to the Band Colours box (bottom right hand side) to get the colours to go from Blue > 80 and Red < 0. So your screen should now look like this;

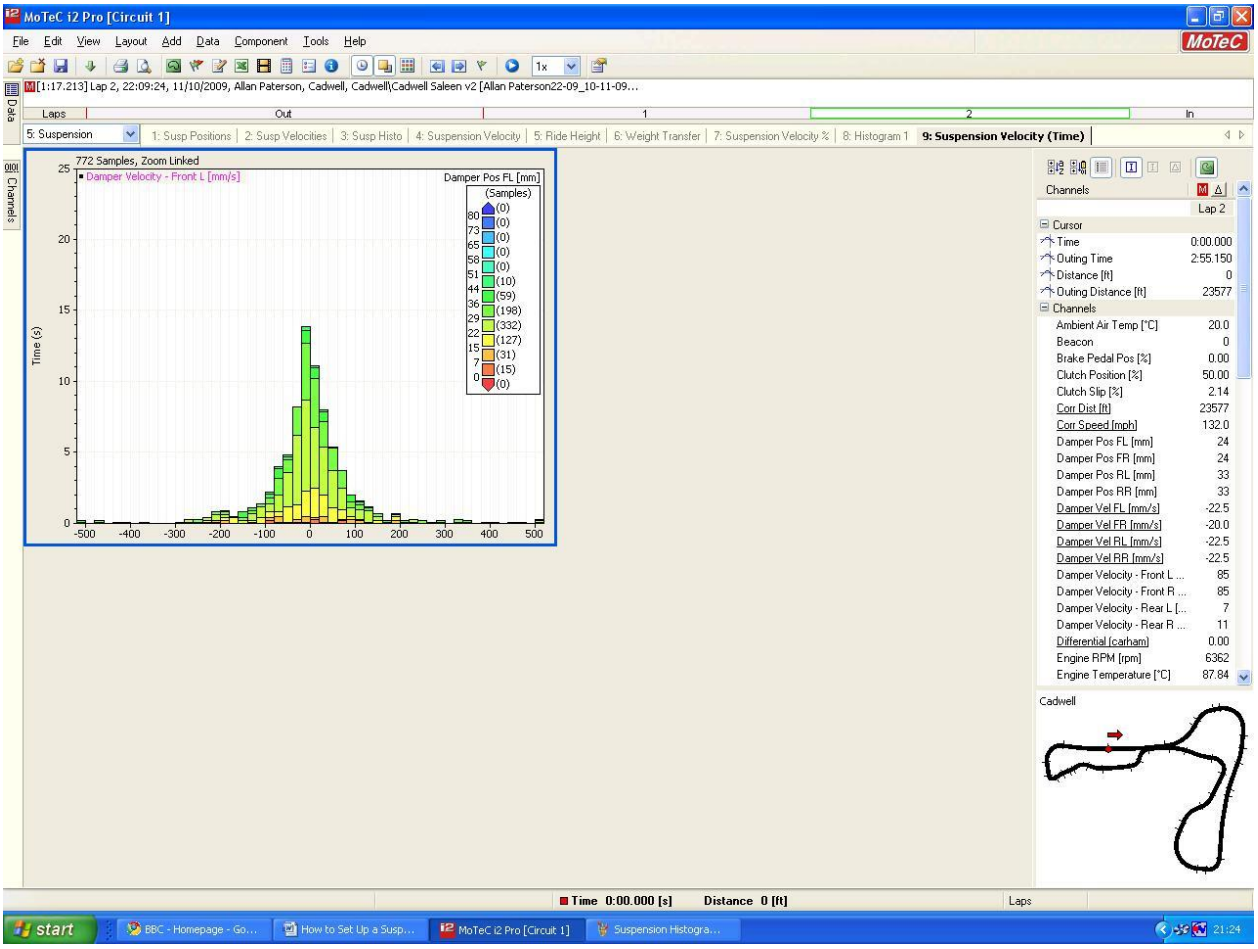
Next Page



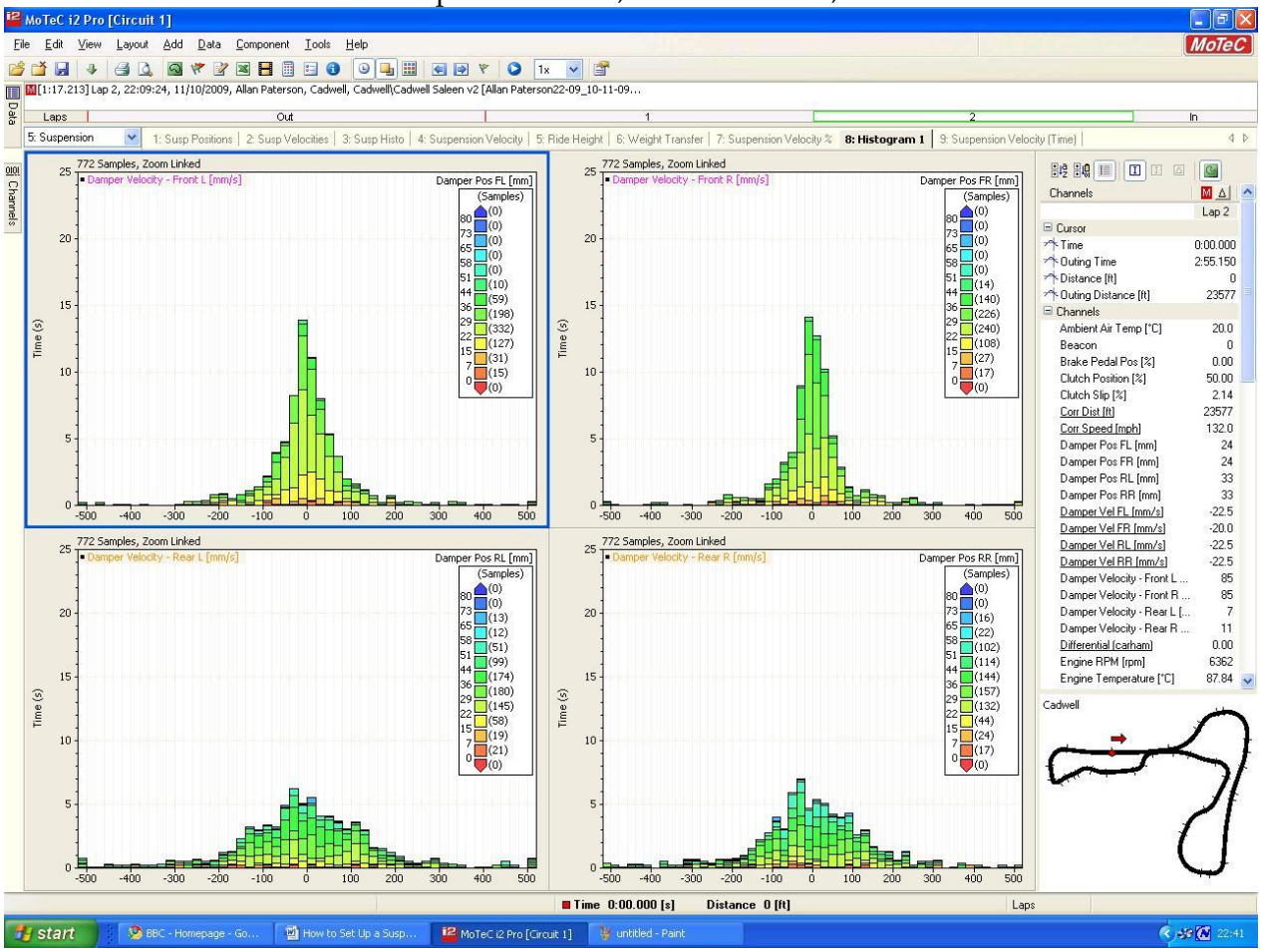
Click on OK and you should get a Finished Screen Like this;



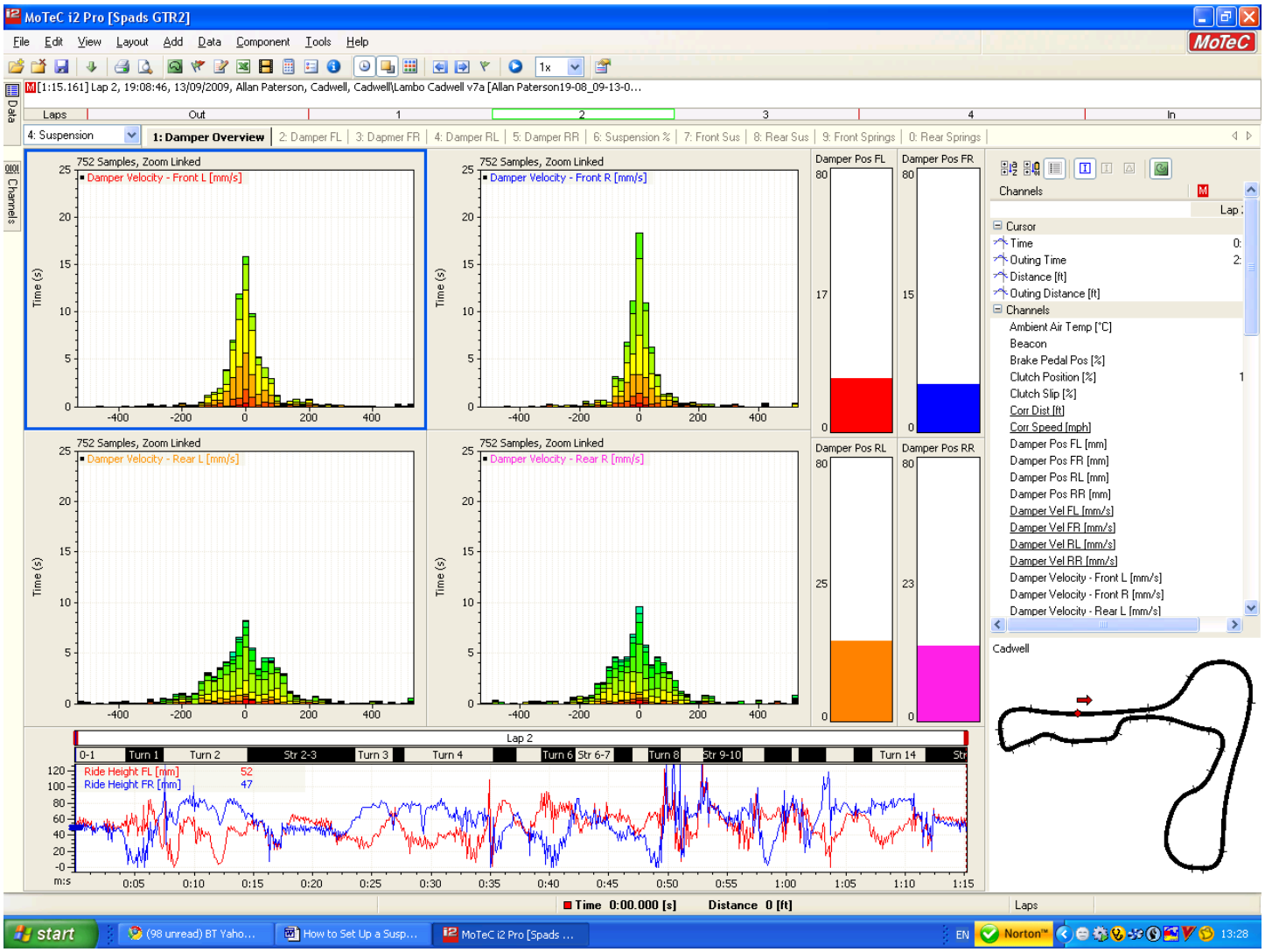
Now you need to hover your mouse over the bottom right hand corner until you get a diagonal double arrow headed cursor, click and hold the corner and drag to resize the window to ¼ of the screen area, it should look like this;



Now you need to repeat the above steps until you have Histograms for all four corners, resizing each to fit the space available, to look like this;



You may chose (personal preference) to modify your set up to look like this;



This involves resizing the Histogram boxes you already have and adding the components in turn or other components that might help you like a time/distance trace, I'll leave that to your personal choice.

Interpreting the Data

The first question to ask is what does this data represent? Well there are three things being displayed; the colours of the bars show you what position (from 0 to 80mm) the damper was in when the reading was taken. If you are using this on different Sims you may well want to adjust the range being graphed – for GTR2 I have experimented and changed the range to 0 to 120mm but you do not get values above 80mm so there is no point.

Secondly each histogram is representing both "Rebound" on Left hand side of zero (negative numbers) and "Bump" on the Right hand side of zero (positive numbers) so each histogram represents two elements of data per wheel. The scale is – 500 to + 500, you can ignore the plus and minus signs that only indicates Bump and Rebound the 0 to 500 part of each side is the speed at which the damper is moving in Bump and Rebound, with a maximum of 500mm/second for each part. 500mm/sec is too "fast" and needs slowing down with increased Damper numbers. However you do not want it too "slow" either, as this means the suspension is not able to absorb the bumps in the road by moving quickly enough, instead the suspension will hit the bumps not absorb them and slow you down.

So in summary, colours = Damper Position, X axis is the Speed of the Damper and the Y axis tells you how much time the damper spent at that speed during the period being analysed (usually one lap).

The histograms themselves should be "cone shaped" higher in the middle and going lower the further away from the centre zero you are on each side. The settings for the histograms should mean Green histograms, with a cone shape, not too wide are a "good set-up". **Note:** This is a general rule and as with all set ups down to personal preference, perfectly good set ups on bumpy tracks, for example, might be wider, flatter more yellow, with only a little green.

Ideally the left hand side of each histogram should be higher (slightly) than the right hand side, if you follow proper suspension set up theory.

At this point I include the text direct from Spad's thread (with permission being sought) for a further explanation;

"The Suspension Histogram is read by looking at the time the spring spends moving. The further away from zero it is the faster it moves and the closer to zero it is the slower it moves. From this you can tell how fast or slow the dampers are controlling the springs. Ideally you want all four wheels running in symmetry.

What you need to do is get a nice cone where the springs are moving at similar speeds. It's all done using damper settings and also by adjusting the spring rates but more so dampers as it is these that control the rate or speed at which the spring moves. Too quick and you will have wider graphs and too slow and they will be very close. These basically equate to over damped (wide) ad under damped (close). Under damped makes the car wallow and feel sluggish and slow on turn entry and exit where as over damped will make the car feel very skittish and unstable over bumps and feel over active on turn entry and exit etc.

Note: I think, Spad got his definitions the wrong way around in this sentence, see my explanation below.

Experiment by setting the dampers as slow (low damper setting) as possible and then go to the other extreme by running them as Fast (high damper settings) as possible. Also try different spring rates to."

I think that explains it well, also later in the thread he wrote;

WIDE = Fast and NARROW = Slow. Again its a balancing act to get them just right to how you like them. I normally aim for about 100'ish milliseconds (ms) in both bump and rebound on the Damper overview but anything over say 200'ish is starting to get too quick and anything under 50'ish ms is getting too slow. You will know when your dampers are set right because you will be able to literally point the car where you want it to go!

So my understanding of what the histograms are telling you is that a WIDE histogram means the suspension is moving too fast and needs to be slowed down by increasing the amount of Damper setting. A NARROW graph is too well damped and therefore needs speeding up by decreasing the amount of damper setting.

The colour change is less relevant and although "green is better" perfectly good set ups will possibly be in other colours, you do however want to dial out the extremes of Red and Blue. Red is bad news as your suspension is bottoming (not the car, the suspension), this condition needs stiffer springs or more Bump dampening. A tiny amount of red on some circuits maybe just okay.

Finally, you need to put Damper Settings in context – it is generally accepted that Damper settings are possibly the last element of set up and used to fine-tune a set up rather than a "fundamental" setting. So, for example you need to get the spring set up as good as you can first, before fine tuning with the damper settings, other wise you are "fine tuning" a fundamentally incorrect set up.

I really did this to help people set up the Histograms in MoTeC rather than give a set up guide, there are plenty of people more qualified than me that can explain it a lot better, including in this Beginners Guide Part II, which you can download from here;

http://www.nogripracing.com/details.php?filenr=22569

A rule of thumb on set ups, as described in the guide, would be to tackle these items in order;

1) Tyre Pressure
2) Camber
3) Gear Ratios
4) Ride Height
5) Brake Balance
6) Steering Lock
7) Castor
8) Toe In/Out
9) Differential Settings
10) Anti Roll Bars
11) Wings
12) Ride Height – and Packers (only if required)
13) Spring Strength
14) Dampers

Of course it is all iterative, in that each change affects everything else to a greater or lesser degree and you will go back and forth fine tuning for sometime before everything is exactly as you want it – that final part (underlined) is quite important what suits you might not suit me and two quite different set ups can deliver the same lap time for two different drivers.

A nice quote from Jenson Button this year (2009), talking about Barrichello's set ups compared to his on the BBC website;

"But I can look at his set-up sheet and I know which bits of it would not work for me. We each might pick and choose a little of what the other one is doing but basically I have to go my own way. If I just copied his set-up, he would be quicker. That's just the way it is."

I've shown this to a couple of people before posting and included their comments in my text but some are worth including verbatim, for example from Carham;

- The histograms can be useful in balancing out the damper reactions to the various vehicle input forces encountered. Or, as you suggest, tailoring the bump and rebound reactions in a non-symmetrically desired way.

- Four corner symmetry may not always provide the "best" set-up. But at least, the histogram can give you a picture of what you're using.

- Damper settings, in my view, are not necessarily the "key" to the set-up, rather they allow a "fine tuning" of the rate of weight transfer onto or off of selected wheels. Or possibly for example, to delay front dive on heavy breaking to minimise initial front camber change. Damper rates are sometimes selected simply to keep the tires in contact with the road surface (bumpy/irregular surfaces). If riding over curbs is mandatory to a quick lap, you will tune the dampers to keep from upsetting the car badly and to keep the tires in contact with the road surface as much as possible. **Having the ability to tune for the "fast" inputs separately from the "slower" inputs, is a very useful feature of modern dampers**....fortunately we have that feature in GTR2.

- Damper settings are usually one of the last set-up details to tune in and may be used to alleviate a secondary issue.

Since getting some great feedback especially from Eagleone there are a couple of other points to make;

I set this up to be the same as Spad's, using Time for the Y Axis. That is fine for short circuits but go somewhere like Le Mans or worse the Nordschleife and the Histogram will get very "full" so you may prefer (if you use those tracks or other long ones) to use Percentage instead for the Y axis.

Now you know how to set it up you will be able to easily change each item to fine-tune it!

You could set automatic scaling but this can make it difficult to understand/compare Front and Rear Histograms (as the data will set different scales) so best stick to 25 seconds (for time) and 25% for percentage. Again for long circuits you may need to modify the values.

Also some of you will have realised that the area under the graph is the total elapsed time for the segment of your data you have chosen. So if you have chosen an exact lap segment the area under the graph is actually your lap time!!

I'll happily include other helpful comments if this generates any further feedback.

Version: 1.1
Date: 23/9/09

Version History

Number	Comment	Date
1.0	Original	20/9/09
1.1	Interpretation info from Eagleone added	23/9/09
2.0	Changed the Histogram plots to 4 individual Histograms as the first method gave incorrect colours As the were all based on the same data	02/04/10